Temporal Emergence from Dynamical Observer Coupling: A Matrix Model for Observer-Dependent Symmetry Breaking

Version 6: Trajectory-Mean Analysis with Corrected Sampler

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Abstract

The Structural Time Framework (STF) proposes that time emerges from observer-dependent ordering of pre-temporal states. We present a computational demonstration using a matrix model where an explicit observer system Y dynamically creates the temporal direction in the main system X, without any external parameters.

Our model employs a directional coupling mechanism where the observer generates a direction vector v from its matrix traces, and dimensions of X aligned with v receive a temporal expansion term $-g_{XY}\hat{v}_\mu^2\mathrm{Tr}(X_\mu^4)$. All results reported here use the **v6 trajectory-mean estimator**: observables are averaged over the measurement window (30 recorded sweeps) rather than read from a single final configuration. Monte Carlo simulations with statistical analysis (5 seeds per configuration, with a 30-seed follow-up for the observer-dimension study) show:

1. Temporal emergence occurs robustly in two one-dimensional scans — observer dimensions $d \in \{2, 3, 4, 5\}$ at fixed $N = 6$, and system sizes $N \in \{4, 5, \ldots, 16\}$ at fixed $d = 3$ — with mean ratios clustering in 4.4–4.8 and alignment at 100% in every configuration. (Pairs such as $d = 5, N = 16$ were not scanned together.)
2. No crossover or two-regime scaling is observed: emergence remains healthy across the entire tested range of N up to $N = 16$, with no threshold at which fixed coupling $g_{XY} = 0.8$ fails to temporalize the system.
3. Perfect alignment between the temporal direction $X_{\max}$ and observer direction $v_{\max}$ in all healthy emergence cases.
4. A follow-up with $n = 30$ seeds per condition shows a weak but detectable monotonic decrease of the mean ratio with observer dimension (slope ≈ -0.08 per unit d, $p = 0.003$, $R^2 = 0.07$), although the pairwise comparison between $d = 3$ and $d = 4$ is not significant at this sample size ($t = 0.52$, $p = 0.61$). The earlier $n = 5$ result ($p = 0.049$) was a small-sample fluctuation.

Important Caveats: This work represents a toy-level mathematical exploration using small matrix sizes ($N \leq 16$). Results demonstrate mathematical consistency, not physical reality.

Keywords: dynamical observer, temporal emergence, matrix models, symmetry breaking, computational demonstration

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1 Introduction

The nature of time has been debated by philosophers and physicists for centuries [8, 1, 10]. The Structural Time Framework (STF) [3] proposes that time is not fundamental but emerges from observer-dependent ordering of pre-temporal states.

In our previous work [2], we presented a toy model where temporal emergence was driven by an external adapter parameter κ_2. While successful, this approach had a fundamental limitation: the observer was represented as a parameter, not as a dynamical system.

This paper addresses that limitation by introducing an **explicit dynamical observer**—a matrix system Y that dynamically creates the temporal direction in the main system X through coupling, without any external parameters.

Important Disclaimer: We emphasize that this work is a **toy-level computational demonstration**, not empirical physics. Our simulations use small matrix sizes ($N \leq 16$), finite Monte Carlo samples, and idealized actions. The results demonstrate mathematical consistency, not physical reality.

2 Dynamical Observer Model

We consider two coupled matrix systems:

Main System X: D Hermitian matrices X^μ of size $N \times N$.

Observer System Y: d Hermitian matrices Y^a of size $N \times N$.

The total action is:

$$S_{\text{total}} = S_X + S_Y + S_{\text{coupling}} \tag{1}$$

where S_X and S_Y are IKKT-like actions with stability gates, and the directional coupling is:

$$S_{\text{coupling}} = -g_{XY} \sum_\mu \hat{v}_\mu^2 \text{Tr}(X_\mu^4) \tag{2}$$

The observer generates direction $\hat{v}$ from its traces. This coupling temporalizes dimensions aligned with $\hat{v}$.

2.1 Theoretical Motivation for the Coupling

The coupling term can be understood from three perspectives:

1. **Effective Field Theory:** Leading-order term in $1/N$ expansion.
2. **Symmetry Breaking:** Explicitly breaks SO(D) $\to$ SO(D-1).
3. **Measurement Analogy:** Observer selects direction, system responds.

2.2 Why the Observer Does Not Temporalize Itself

The distinction between observer and system is relational, not fundamental. The asymmetry in coupling reflects the measurement paradigm: the apparatus measures the system, not vice versa. This avoids the self-observation paradox analogous to Wigner's friend.

3 Computational Methodology

We use Metropolis-Hastings sampling [7] with:

- Matrix size: $N \in \{4, 5, 6, 7, 8\}$, with a larger-N extension to $N \in \{9, 10, 12, 14, 16\}$ (Section 4.3)
- System dimensions: $D = 6$
- Observer dimensions: $d \in \{2, 3, 4, 5\}$
- Coupling strength: $g_{XY} \in [0.8, 1.15]$
- Thermalization: 20 sweeps
- Measurement: 30 sweeps
- Multiple seeds: $\{42, 43, 44, 45, 46\}$ (base set), plus $\{100, \ldots, 129\}$ for the 30-seed observer-dimension study

Thermalization check: the observables reported below are robust to increasing the thermalization budget. Re-running representative configurations with $n_{\text{therm}} = 20, 40, 80, 160$ leaves the mean ratio and alignment unchanged (e.g. $R \approx 4.72$–4.84 at $N = 8$, $d = 3$, $g_{XY} = 0.8$), and the wall fraction stays at zero throughout. We therefore regard $n_{\text{therm}} = 20$ as sufficient for the healthy (v6) regime studied here. (For the separate two-way-coupling model of v7, in which the observer itself can be driven into the stability gate, $n_{\text{therm}} = 20$ was found insufficient and a larger budget is required; that model is not reported in this paper.)

The sampler is the **correctness-first v6 engine** (`code/sgoed_core_v6.py`). It computes Metropolis acceptance via an incremental local-energy delta rather than a full-action recomputation; the delta was validated against the full action to machine precision (maximum relative error $< 10^{-9}$ over 4000 randomized trials, including wall-regime edge cases). The v6 sampler updates every matrix element at every sweep. This replaces the earlier v5 engine, which applied a `step=2` element-skip optimization for $N > 6$ (updating only even-indexed matrix elements)—a shortcut that we subsequently identified as the source of a spurious finite-size crossover and a spurious variance spike at $N = 8$ (Section 4.5).

All observables are recorded over the measurement window and then averaged (**trajectory mean**), rather than read from a single final configuration.

Observables measured, with $\text{extent}(X_\mu) \equiv \text{Tr}(X_\mu^2)/N$:

$$\text{Ratio} = \frac{\text{extent}(X_{\max})}{\text{mean}(\text{extent}(X_{\neq\max}))} \tag{3}$$

$$\text{Alignment} = \mathbb{I}(\arg\max_\mu \text{extent}(X_\mu) == \arg\max_\mu \hat{v}_\mu^2) \tag{4}$$

$$H = -\sum_\mu p_\mu \ln p_\mu, \quad p_\mu = \frac{\text{extent}(X_\mu)}{\sum_\nu \text{extent}(X_\nu)} \tag{5}$$

A configuration is classified as **Healthy** if the mean Ratio over the sampled seeds satisfies $1.5 < \overline{R} < 10.0$; configurations with $\overline{R} \leq 1.5$ are labeled Weak. The upper bound $R = 10$ is chosen to match the stability-gate cutoff on the per-matrix *extent* (`max_extent`= 10.0) used in the action itself. We emphasize that the two quantities are distinct: the gate acts on extent$(X_\mu) = \mathrm{Tr}(X_\mu^2)/N$ of a *single* matrix (a dimensionful scale), whereas R is the *dimensionless ratio* of the largest extent to the mean of the remaining extents. When the aligned dimension is driven toward the gate ceiling, its extent saturates just below 10.0 while the others remain near 1, so R approaches ~ 10 even though no individual extent ever exceeds the cutoff (the gate penalty suppresses growth before the threshold is crossed). None of the configurations reported below have an individual extent exceeding 10.0.

4 Results

4.1 Statistical Robustness

Table 1 summarizes results for optimal parameters.

Table 1: Statistical analysis ($N = 6$, $g_{XY} = 0.8$)

d	Mean Ratio	Std Dev	CV (%)	Alignment
3	4.65	± 0.23	4.9%	100% (30/30)
4	4.61	± 0.32	7.0%	100% (30/30)

A follow-up with $n = 30$ seeds per condition (rather than the initial $n = 5$) gives a paired t-test of $t = 0.52$, $p = 0.61$ for the $d = 3$ versus $d = 4$ comparison. This is not significant. The marginally significant result reported in an earlier draft of this version ($t = 2.804$, $p = 0.049$ at $n = 5$) is therefore attributed to a small-sample fluctuation and is withdrawn: with a larger sample the pairwise difference between $d = 3$ and $d = 4$ is not resolved.

Across all four observer dimensions ($d = 2, 3, 4, 5$, $n = 30$ each), however, the mean ratio decreases monotonically from 4.74 ($d = 2$) to 4.49 ($d = 5$), a linear trend of slope ≈ -0.08 per unit d ($p = 0.003$, $R^2 = 0.07$). The trend is weak—it explains only $\sim 7\%$ of the seed-to-seed variance—but is statistically detectable at this sample size, consistent with a mild "dimensional dilution" of the directional coupling. It should not be read as evidence that any single pair of dimensions differs.

4.2 Observer-Dimension Phase Diagram

To investigate the role of observer complexity, we varied the observer dimension $d \in \{2, 3, 4, 5\}$ while keeping $N = 6$, $D = 6$, and $g_{XY} = 0.8$ fixed. Each configuration was evaluated over five independent random seeds.

Several features are noteworthy.

1. **No strict threshold at $d = 3$:** Contrary to our initial expectation, even $d = 2$ produces healthy temporal emergence with perfect alignment. This suggests that the minimum observer complexity required for temporalization is lower than anticipated.

2. **Robust emergence, with a weak downward trend in d:** Mean ratios cluster in a narrow range (4.49–4.74) across all four tested observer dimensions, and alignment is perfect (100%, 30/30 seeds) in every case. With $n = 30$ seeds per dimension the

Table 2: Phase diagram by observer dimension ($N = 6$, $g_{XY} = 0.8$)

d	Mean Ratio	Std Dev	CV (%)	Alignment	Status
2	4.74	± 0.15	3.1%	100% (30/30)	Healthy
3	4.65	± 0.23	4.9%	100% (30/30)	Healthy
4	4.61	± 0.32	7.0%	100% (30/30)	Healthy
5	4.49	± 0.46	10.3%	100% (30/30)	Healthy

mean ratio decreases monotonically from 4.74 ($d = 2$) to 4.49 ($d = 5$), a weak trend (slope ≈ -0.08 per unit d, $p = 0.003$, $R^2 = 0.07$; Table 1); the paired comparison between $d = 3$ and $d = 4$ is, however, not significant at this sample size ($p = 0.61$). We therefore report a mild "dimensional dilution" of the directional coupling rather than a robust pairwise difference, and we find no evidence for an optimal or "sweet spot" observer dimension in this range. (An earlier v5 draft of this table reported a spurious $d = 4$ "sweet spot" and reduced $d = 5$ alignment; those numbers could not be reproduced from the corrected v6 sampler and have been replaced with freshly re-run, seed-logged values.)

4.3 Finite-Size Scaling and Parameter Sensitivity

We next tested whether the observed temporal emergence is a finite-size artifact by varying the matrix size N. We first fixed $d = 3$ and $g_{XY} = 0.8$ and examined $N \in \{4, 5, 6, 7, 8\}$.

Table 3: Finite-size scaling with fixed $g_{XY} = 0.8$ ($d = 3$)

N	Mean Ratio	Std Dev	CV (%)	Alignment	Status
4	4.63	± 0.19	4.2%	100% (5/5)	Healthy
5	4.62	± 0.18	4.0%	100% (5/5)	Healthy
6	4.75	± 0.17	3.5%	100% (5/5)	Healthy
7	4.75	± 0.12	2.6%	100% (5/5)	Healthy
8	4.72	± 0.05	1.0%	100% (5/5)	Healthy

To test whether this robustness is a small-N accident, we extended the scan to $N \in \{9, 10, 12, 14, 16\}$ (same $d = 3$, $g_{XY} = 0.8$, five seeds each):

Table 4: Larger-N extension with fixed $g_{XY} = 0.8$ ($d = 3$)

N	Mean Ratio	Std Dev	CV (%)	Alignment	τ_{int}
9	4.75	± 0.10	2.0%	100% (5/5)	1.8
10	4.77	± 0.06	1.3%	100% (5/5)	2.2
12	4.76	± 0.10	2.1%	100% (5/5)	1.8
14	4.75	± 0.03	0.6%	100% (5/5)	1.6
16	4.72	± 0.07	1.6%	100% (5/5)	1.9

The robustness extends to $N = 16$: the ratio remains at $R \approx 4.7$–4.8, alignment stays perfect, and the wall fraction is zero throughout. The integrated autocorrelation time

$\tau_{\text{int}} \approx 1.6$–$2.2$ (measurement sweeps) confirms that the 30 recorded trajectory points are only weakly correlated, so the trajectory-mean estimates are well behaved.

In contrast to an earlier v5 analysis, the corrected sampler shows **no finite-size crossover**: the ratio remains healthy and statistically stable across the entire tested range $N = 4$–16.

1. **No crossover regime:** At every tested size the mean ratio stays within 4.6–4.8 and alignment is perfect (100%). There is no N at which fixed coupling $g_{XY} = 0.8$ fails to produce temporalization.

2. **CV decreases monotonically with N:** The coefficient of variation falls steadily from 4.2% at $N = 4$ to about 1% at $N = 8$ and remains at or below $\sim 2\%$ through $N = 16$. Larger systems produce more tightly self-averaging measurements, consistent with the trajectory-mean estimator averaging over more matrix degrees of freedom.

The apparent crossover at $N \approx 7$ reported in v5 (ratio dropping to ≈ 1.8 with reduced alignment) is not reproduced here. As documented in Section 4.5, it was an artifact of the v5 `step=2` element-skip shortcut, not a feature of the model.

4.4 Parameter Tuning at $N = 8$

To examine how the coupling strength affects the strong-coupling side of the model, we repeated the $N = 8$ simulations while varying g_{XY}.

Table 5: Parameter tuning at $N = 8$ ($d = 3$)

g_{XY}	Mean Ratio	Std Dev	CV (%)	Alignment	Status
0.80	4.72	±0.05	1.0%	100% (5/5)	Healthy
1.05	10.09	±0.20	2.0%	100% (5/5)	Gate ceiling
1.10	10.22	±0.04	0.4%	100% (5/5)	Gate ceiling
1.15	10.34	±0.23	2.3%	100% (5/5)	Gate ceiling

The corrected sampler changes the picture materially. At $g_{XY} = 0.80$ the system is already healthy ($R \approx 4.72$, CV 1.0%), so **no parameter tuning is required at $N = 8$.** For $g_{XY} \geq 1.05$ the ratio saturates near $R \approx 10$ with CV near 1–2%; this is the gate ceiling described in Section 3: the aligned dimension's *extent* is driven to just below the `max_extent`= 10.0 cutoff (individual extents peak at ≈ 9.99 and never exceed 10.0), so the dimensionless ratio approaches ~ 10 while the gate penalty suppresses further growth.

Two points follow. First, the non-monotonic CV spike at $g_{XY} = 1.05$ (CV = 59.0%) reported in v5 is not reproduced here; CV is flat and small (0.4–2.3%) across the scanned range. Second, $g_{XY} = 0.80$ is the preferred working value at $N = 8$: it produces healthy emergence well inside the Healthy window, and increasing g_{XY} beyond ~ 1.05 only pins the aligned extent against the gate without further benefit.

4.5 On the Absence of a Crossover Regime

Unlike our earlier v5 results (which used a final-snapshot estimator and a `step=2` element-skip shortcut for $N > 6$), the v6 trajectory-mean results show **no evidence of a crossover or phase transition** in the range $N = 4$–16. The ratio remains stable at $R \approx 4.6$–4.8 across all tested system sizes, with alignment 100% and CV *decreasing* with N. This demonstrates that the apparent crossover at $N \approx 7$—and the accompanying "two-regime scaling" with a tuned optimal coupling $g_{XY}^{\text{opt}}(N)$—was an artifact of the v5 sampling strategy, not a feature of the model.

Specifically, the v5 engine applied

```
step = 2 if N > 6 else 1
```

which caused the X-matrices at $N = 7, 8$ to have only their even-indexed elements updated, leaving the odd-indexed elements frozen near their initial values and preventing the system from thermalizing. The Y-matrices were not subject to this skip, so the two subsystems were treated asymmetrically. The v6 engine removes this shortcut and updates every element at every sweep; once the system is allowed to thermalize fully, the finite-size "crossover" disappears and the mechanism is seen to be robust across the entire tested range.

We therefore withdraw the earlier two-regime scaling picture and the power-law estimate $\alpha \approx 0.46$–1.10 for $g_{XY}^{\text{opt}}(N)$. There is no longer a set of tuned optima from which such an exponent could be computed: $g_{XY} = 0.8$ suffices for every $N \in \{4, 5, \ldots, 16\}$.

4.6 Physical Interpretation: Robustness and Gate Saturation

With the crossover removed, the two physically meaningful observations are the *robustness* of temporalization across N and d, and the *saturation* of the aligned extent against the stability gate at strong coupling.

1. **Robustness across system size:** The mechanism produces a healthy ratio and perfect alignment at every tested size, with no required change in coupling. This is consistent with the coupling energy remaining sufficient relative to thermal fluctuations throughout $N \leq 8$; the decrease of CV with N further indicates that larger systems self-average, rather than destabilizing.

2. **Gate saturation at strong coupling:** For $g_{XY} \gtrsim 1.05$ at $N = 8$, the aligned dimension's extent is driven to the stability-gate ceiling (≈ 9.99, just below `max_extent`= 10.0). The gate penalty term $10.0\,(\text{extent} - 10.0)^2$ then balances the coupling, pinning the extent just below the cutoff. The resulting ratio approaches ~ 10 while no individual extent ever exceeds 10.0; this is the gate doing its job, not a new physical phase.

3. **Spectral mechanism (rank-1 condensation):** The directional coupling acts on the fourth moment $\text{Tr}(X_\mu^4)$, which strongly favors states in which a single eigen-direction dominates. At $g_{XY} = 0.8$, $N = 8$, the selected matrix is effectively rank-1: one eigenvalue $\lambda_{\max} \approx 6$ with the next-largest eigenvalue an order of magnitude smaller ($\lambda_{\max}/\lambda_2 \approx 20$), whereas in the uncoupled baseline ($g_{XY} = 0$) the spectrum remains dispersed ($\lambda_{\max}/\lambda_2 \lesssim 1.3$). The observed ratio (4.5–4.7) is therefore not an artifact of the observable: it signals genuine eigenvalue condensation driven by the coupling, separated from the $g_{XY} = 0$ baseline (1.10 ± 0.05) by well over fifty standard deviations.

4. **Weak observer-dimension trend:** Mean ratio decreases gently with observer dimension ($4.74 \to 4.49$ from $d = 2$ to $d = 5$ at $n = 30$), suggesting that additional observer degrees of freedom dilute the directional coupling slightly. The trend is statistically detectable (slope ≈ -0.08 per unit d, $p = 0.003$) but weak ($R^2 = 0.07$), and no single pair of dimensions differs significantly.

4.7 Connection to IKKT Matrix Model

Our model shares the mechanism of directional symmetry breaking with the IKKT matrix model [5, 6], in which Kim and Nishimura found that spontaneous symmetry breaking SO(9) $\to$ SO(3) requires sufficiently large N for clear emergence of (3+1)-dimensional spacetime. We note, however, that the corrected v6 results do *not* exhibit the finite-size sensitivity that an earlier v5 analysis had suggested: within the tested range $N \leq 8$, our coupling $g_{XY} = 0.8$ produces robust emergence with no required N-dependent tuning. Any comparison between the two models therefore rests on the shared qualitative idea of competition between coupling energy and the phase space of matrix configurations, not on a demonstrated finite-size crossover in our model.

5 Discussion

5.1 Advantages Over External Adapter

The dynamical observer approach offers several advantages:

1. **Physical Motivation:** Observer is dynamical system, not parameter.
2. **Emergent Direction:** Temporal direction emerges from observer dynamics.
3. **Robustness:** Works across $d = 2, 3, 4, 5$ and all tested $N = 4$–16 with fixed coupling, no crossover and no required parameter tuning.
4. **Measurement Theory:** Observer as measurement device.

5.2 Relation to Philosophical Positions

Our model is consistent with several philosophical positions within the restricted scope of our toy model:

- **Leibniz (relational time):** Temporal direction created by observer.
- **Barbour (timeless physics) [1]:** Pre-observer state is symmetric.
- **Rovelli (thermal time) [4, 10]:** Coupling creates preferred direction.
- **Relational QM [9]:** States relative to observer.

Important: "Consistent with" does not mean "proves" or "validates." These philosophical positions remain open questions.

6 Open Questions and Future Directions

Our results raise several important questions that warrant further investigation:

1. **Gate saturation mechanism:** At $N = 8$ and $g_{XY} \gtrsim 1.05$ the aligned extent saturates against the stability-gate ceiling, pinning the ratio near 10. A fine scan of g_{XY} in the interval 0.80–1.05 (step 0.025) shows the entry into saturation is a *smooth crossover*, not a sharp transition: the ratio rises monotonically from ≈ 4.7 to ≈ 10, with the steepest single step ($\Delta R \approx 1.7$) near $g_{XY} \approx 0.875$–0.90, and then flattens onto the plateau for $g_{XY} \gtrsim 0.95$. Whether the saturation value is a well-defined function of `max_extent` remains open.

2. **Large-N behavior:** The corrected sampler shows robust emergence with fixed coupling up to $N = 16$ (Section 4.3, Table 4). Whether this robustness persists at still larger N ($N = 18, 20, \ldots$) remains open; a genuine finite-size crossover could still emerge beyond the currently tested range, and would then need to be distinguished from the v5 artifact described in Section 4.5.

3. **Observer-dimension dependence (dimensional dilution):** With $n = 30$ seeds per dimension, the mean ratio decreases monotonically from 4.74 ($d = 2$) to 4.49 ($d = 5$). The trend is statistically detectable (slope ≈ -0.08 per unit d, $p = 0.003$) but weak ($R^2 = 0.07$), and the pairwise comparison between $d = 3$ and $d = 4$ is not significant ($p = 0.61$). This is consistent with a mild "dimensional dilution" of the directional coupling over additional observer directions, rather than any sharp dependence. Confirming the physical origin of the dilution (and the concomitant rise of CV with d, from 3.1% at $d = 2$ to 10.3% at $d = 5$) would require still larger samples or an analytic argument.

4. **Universality:** Would this mechanism work with different coupling structures or different matrix models (e.g., BFSS, Lorentzian IKKT)? How universal is the observer-induced temporalization phenomenon?

5. **Feedback coupling (back-reaction):** The present model treats the observer Y as an external field acting on X with no back-reaction. To move beyond this one-way picture, one could introduce a small back-reaction term from X back to Y (e.g. a g_{YX}-weighted coupling in the observer action) and ask whether the two-way coupled system still selects a stable temporal direction, or whether the mutual feedback destabilizes the ordering. This is a natural step toward a genuinely relational, rather than externally driven, notion of time.

7 Conclusion

We have presented a toy-level computational demonstration of temporal emergence from a dynamical observer. Our key findings:

1. An observer system Y can dynamically create the temporal direction in a main system X through directional coupling.

2. With the corrected v6 trajectory-mean sampler, we observe healthy temporal emergence (ratios 4.4–4.8, alignment 100%) across the entire tested range: observer dimensions $d \in \{2, 3, 4, 5\}$ and system sizes $N \in \{4, 5, \ldots, 16\}$, with fixed coupling $g_{XY} = 0.8$.

3. No finite-size crossover or two-regime scaling is observed; the crossover and tuned-coupling picture reported in v5 was an artifact of a `step=2` sampling shortcut, now removed (Section 4.5).

4. Perfect alignment ($X_{\max} = v_{\max}$) in all healthy emergence cases confirms the observer's active role.

5. At strong coupling ($g_{XY} \gtrsim 1.05$, $N = 8$) the aligned extent saturates against the stability gate, pinning the ratio near 10; this is gate saturation, not a new phase.

6. Across observer dimensions, the mean ratio decreases weakly and monotonically with d (slope ≈ -0.08 per unit d, $p = 0.003$, $R^2 = 0.07$ at $n = 30$); the pairwise comparison between $d = 3$ and $d = 4$ is not significant ($p = 0.61$). The earlier $n = 5$ "marginally significant" result ($p = 0.049$) was a small-sample fluctuation and has been withdrawn.

Note on the v5 → v6 correction: An earlier version of this manuscript (v5) reported a finite-size crossover at $N \approx 7$, a two-regime scaling picture with a tuned optimal coupling $g_{XY}^{\text{opt}}(N)$, and a non-monotonic variance spike at $g_{XY} = 1.05$. None of these are reproduced by the corrected v6 sampler. They were traced to a `step=2` element-skip optimization in the v5 engine, which for $N > 6$ updated only even-indexed matrix elements and prevented full thermalization. The v5 numbers have therefore been withdrawn and replaced by the v6 trajectory-mean results reported here; the v5 manuscript is retained unchanged as `manuscript_v5.tex` for archival comparison.

Appropriate Interpretation: These results demonstrate that dynamical observer temporalization *can be represented* in a consistent mathematical framework. They do not prove that time emerges from observer coupling in nature.

Final Statement: This work is a toy-level demonstration of mathematical consistency, not a discovery of physical laws. Following the philosophy of our exploratory notes, we emphasize that this is a **design framework demonstrating mathematical consistency**, not empirical physics.

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